



P0

Structure Is the Intelligence

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Structure Is the Intelligence

Working Title (EN): Structure Is the Intelligence: Spatial Organization as the Substrate-Invariant Core of Intelligent Systems

Flagship synthesis paper P0 of the whole volume • UDOS Reasoning Engine v7.5.0 • 2026-09-19 This paper is the master outline of the 12 topical papers (P1–P12). It adds no independent experiment; rather, it converges the measured evidence scattered across the topical papers and the established knowledge spanning the four natural-science chapters onto one falsifiable central claim. Evidence discipline: UDOS figures all trace back to the raw JSON in reports7/, marked verified / cpu-proto / cpu-proxy / unverified; single-seed pilots are written as "pilot point estimates." Natural-science classics were verified one by one on 2026-09-19, and those that could not be verified are marked [CITATION NEEDED].

Abstract

We advance a restrained, falsifiable central claim: **given fixed components (materials, weights) and a fixed energy budget, the intelligent behavior a system exhibits is primarily determined by its spatial relational structure; structure is the substrate-invariant core across chemical bonds, carbon chains, proteins, neurons, and artificial machines or agent collectives, while material and substrate are replaceable implementations.**

We define "structure" strictly as a triple: connectivity topology (who connects to whom), geometry/spatial organization (metric and orientation of relations), and modular boundaries (hierarchy and encapsulation), and distinguish it from "parameters" (weights, material composition) and "dynamics" (energy, learning rules). Around this definition we state four falsifiable predictions, each with an explicit refutation condition. Evidence comes from two roads: a natural-history road (isomerism, protein denaturation, connective lesions) and an engineering road measured on the UDOS engine (star-vs-layered topology, multi-view pose structure, hybrid embodied control, and the unidentifiability of dynamics when structural information is insufficient). We do not claim structure is sufficient; Chapter 6 fully voices the objections. Structure is not everything, but it is the fourth axis—structure axis—systematically under-counted by the prevailing parameter–data–compute scaling narrative.

Keywords: structure; connectivity topology; spatial organization; modular boundaries; substrate invariance; falsifiable prediction; fourth scaling axis; UDOS; multi-agent; embodied intelligence

1 Introduction

1.1 A fact repeatedly seen yet rarely named

Across different scales, "intelligence" repeatedly points to the same non-material thing. A protein's catalytic power is determined not by what its amino-acid atoms are but by the three-dimensional shape those atoms fold into; DNA can replicate itself not because it is made of carbon, hydrogen, oxygen, nitrogen, and phosphorus (countless molecules are made of the same elements) but because it is a double helix wound by complementary base pairing; the human brain can remember and plan not in the firing of a single neuron but in the connectivity among its eighty billion neurons. Placing these facts

together, a simple question surfaces: **if the material is changed while the relational structure is preserved, does the function remain?**

This paper elevates that question into a researchable, falsifiable proposition and argues: separating "structure" from "materials/parameters" and from "dynamics/rules" is an overlooked key to understanding intelligent systems, whether biological or artificial.

Boundaries must be drawn first in the introduction. This paper does not use undefined big words like "universe," "ultimate," or "first principles." The "structure" this paper speaks of is a triple that can be operationalized, measured, and broken in Chapter 2; the "intelligent behavior" this paper speaks of refers to the coordination, prediction, generalization, and control a system exhibits on a given task, not any metaphysical "consciousness." The argumentation strategy is restrained: the introduction and conclusion only state the theme, while the full weight of the body is pressed onto evidence and formalization.

1.2 Why raise this now

Over the past decade, the success of large models has been almost monopolized by one narrative: treating parameter count, data volume, and compute as three coordinate axes, with performance falling along a power law (Kaplan et al., 2020; Hoffmann et al., 2022, i.e. Chinchilla). This narrative is extremely successful, but it has a systematic blind spot: it assumes "what the network looks like" is fixed—all use the Transformer, all stack layers, all stack width, all connect by some established attention pattern—and then only adds parameters along this fixed structural track. Once "structure" is also made a designable variable, a hidden dimension emerges:

- With the same parameters, switching to sparse routing (MoE / Switch, Fedus et al., 2021) multiplies parameter capacity several-fold without increasing activated compute;
- With the same number of agents, a star center's fan-in explodes linearly with scale, while a layered tree can press each node's fan-in to a constant;
- With the same monocular image, adding multi-view pose constraints can raise the coverage of novel-view prediction from 0.7456 to 0.9957.

These phenomena lie on none of the three axes of parameters–data–compute; they occur in the dimension between the axes: **how you connect the nodes, how you arrange the information, how you layer the modules**. This paper names this dimension the structure axis and argues it is the "fourth axis" orthogonal to the three axes.

1.3 Position and contributions of this paper

This paper is the flagship synthesis paper of the whole volume, at a standard above the other 12 topical papers. Its contribution is not to run one more experiment but to:

1. **Give a falsifiable central claim and four explicit refutation conditions** (Chapter 2), converting the correct but empty phrase "structure matters" into a scientific claim that can be killed by experiment;
2. **Connect natural-science and engineering evidence** (Chapters 3, 4), placing the four-chapter natural history of chemical bonds–carbon chains–proteins–neurons and the measured figures of the UDOS engine together on one master table;
3. **Propose a unified framework** (Chapter 5): the structure–dynamics–energy ternary model and a "theorem-like statement" of cross-substrate isomorphism;

4. **Honestly face the objections** (Chapter 6), fully listing why structure is not a sufficient condition and when "structure" degenerates into an unfalsifiable all-purpose explanation;
5. **Give an executable research program** (Chapter 7), marking the gate roadmap from CPU prototypes to GPU and real data.

1.4 Reading route

Chapter 2 is the skeleton of the whole paper and should be read even by non-specialist readers—it defines what exactly we are talking about. Chapters 3 and 4 are evidence and readers may choose by interest. Chapter 5 converges the evidence into a framework. Chapter 6 is this paper's self-defense and the landing point of academic honesty. Chapters 7 and 8 face the future. Chapter 9 lists threats and validity, and Chapter 10 closes. The appendix gives reproduction indices for all measured figures.

2 Terminology and formalization: what "structure" means and what it is not

2.1 The triple definition

This paper defines a system's **structure** as the triple $S=(T, G, M)$:

- **ST connectivity topology**: which units in the system have connections, and the direction and type of connections. It answers "who relates to whom." It does not care how strong a connection is, only whether it connects.
- **SG geometry/spatial organization**: the metric and orientation of connections in an embedded space—distance, pose, angle, the numerical values of fan-in and fan-out. It answers "with what strength, in what direction, at what position things relate."
- **SM modular boundaries**: into which submodules the system is divided, how levels nest, and where the interfaces and encapsulation boundaries between modules lie. It answers "which part counts as a whole."

Figure 1 Conceptual diagram of the structure triple

Figure 1 Conceptual schematic of the structure triple. The three sub-dimensions—connectivity topology, geometry/spatial organization, modular boundaries—together constitute "structure" and are clearly distinguished from "parameters" and "dynamics" below. Schematic, not measured data.

These three sub-dimensions must coexist; missing any one does not constitute complete "structure." A pure graph (only ST , no distances or levels) is insufficient to describe a protein: two different folded chains, even if their connection order (topology) is the same, are different isomers with different functions if their geometric arrangement in 3D space differs. Conversely, a system without modular boundaries—flattening all units into an undifferentiated net—cannot be hierarchically designed or verified even if topology and geometry are clear.

2.2 Strict distinction: structure $\neq$ parameters $\neq$ dynamics

The whole argument of this paper rests on the strict separation of three concepts. This separation is the premise of falsifiability—only when we can fix two and change the third is "structure matters" a testable proposition.

- **Parameters:** connection strengths, the chemical composition of matter, model weights. It is "how strongly things connect," "with what material they connect." Parameters are replaceable, learnable, and can be added or removed.
- **Dynamics:** how signals propagate over connections, how energy flows, how learning rules modify connections. It is "what rules run over the connections."
- **Structure:** whether connections exist ($\$T\$$), the metric and orientation of connections ($\$G\$$), how modules are divided ($\$M\$$). It is "the skeleton of connections itself."

The relation among the three can be given by an intuitive analogy: **structure is the topology and spatial wiring of a circuit diagram, parameters are the resistance values and materials of the components, dynamics is how the current flows and how the components age once powered on.** You can replace copper wires with silver wires (change material/parameters) and raise the voltage (change energy/dynamics), but if you change the circuit diagram itself—for example replacing a bridge circuit with a series one—the circuit's function changes even if the components are identical. This distinction is used repeatedly below, especially in the controlled comparisons of Chapter 4 where "only topology changes, parameters and compute fixed."

It is worth specifically noting that deep-learning research history has long had a tradition of conflating the three. Early work mixed layer count, width, and activation functions together under "architecture," treating "architecture search" and "hyperparameter tuning" as the same thing; when attention was proposed, people first thought it was a new "parameterization method," only later gradually seeing that what it really changes is the topology of "whether and how any two points connect." This paper argues for unpacking this historical conflation: when a researcher says "changing the architecture made it stronger," one must further ask—did they change connectivity topology $\$T\$$, embedded geometry $\$G\$$, or modular boundaries $\$M\$$? If none of the three can be stated, then "architectural progress" remains at the stage of engineering metaphysics. The three-way terminology of this paper exists precisely to make such questioning operational. One more point: between parameters and structure there is an often-overlooked intermediate state—the **existence of a connection can be learned parameterically** (attention weights are "dynamically determined connection topology"), which is precisely where deep learning advances beyond hand-built circuits; but this only shows $\$T\$$ can be shaped by $\$D\$$, not that $\$T\$$ and $\$D\$$ are the same thing. Section 6.4 returns to this coupling.

2.3 The central claim

Central claim (C): given fixed components and a fixed energy budget, the intelligent behavior a system exhibits is primarily determined by its spatial relational structure $\$S=(T,G,M)\$$; $\$S\$$ is the invariant across substrates (molecules–cells–neurons–machines–agent collectives), and materials and substrates are replaceable implementations.

Stated more operationally: there exists a class of tasks in which two systems $\$A\$$ and $\$B\$$ have essentially the same parameter count, material composition, and energy consumption, the only difference being $\$S_A \neq S_B\$$; if $\$A\$$ and $\$B\$$ show order-of-magnitude differences in task performance, and that difference cannot be remedied by adding parameters to $\$B\$$, then claim C is supported.

This is a strong claim, but it is strictly constrained by the refutation conditions of the four predictions below.

2.4 Four falsifiable predictions

The value of a scientific proposition lies not in how much it can explain but in what can kill it. For each of the four predictions below, "what experimental result would refute this paper" is explicitly stated.

Prediction 1 (topology trades for capability, same parameters and compute)

Statement: in two systems with exactly the same total parameters and total compute budget, changing only the connectivity topology $\$T\$$ (for example replacing a star with a layered tree, or with a centerless swarm) produces order-of-magnitude differences in coordination capability, and such differences do not disappear with "larger parameters."

UDOS measured prototype (reports7/matrix_scale_demo.json, cpu-pto): at million-level agent scale (target 1,000,000, actual 2,396,745 units), the star center's fan-in reaches **20,971,520** and the critical-path rounds reach **8,388,608**; the layered tree presses each node's maximum fan-in constantly to **9** and the critical-path rounds to only **14**. Parameter count and task volume are exactly identical; only topology differs.

What result would refute Prediction 1: if in strictly same-parameter, same-compute GPU/real-machine comparisons, the coordination success rate and convergence time of star and layered on the same batch of tasks fall in the same order of magnitude (for example fan-in differs by three orders of magnitude while success rate differs by less than 10%), and that conclusion stably reproduces across multiple seeds and task families, then "topology causes order-of-magnitude capability differences" is falsified, and this paper degrades to the weaker claim that "topology only affects efficiency, not capability."

Prediction 2 (breaking structure while preserving parameters; adding parameters cannot compensate)

Statement: artificially breaking structure (randomly rewiring edges, scrambling view poses, erasing the handoff bundle) while keeping parameter count fixed significantly degrades system performance; at that point simply increasing parameters cannot restore performance.

UDOS measured prototype (the estimator_param_error section of reports7/v7_verification.json, main gate verified, estimator a cpu-pto pilot point estimate): when structural information is insufficient (the observation window too short, equivalent to "structure broken"), the identifiability skill for the dynamics parameter acceleration $\$a\$$ is **-0.508**—worse than predicting with the mean; when structure (window/kinematic structure) is sufficient, the skill for the same type of parameter rises to +0.82 (other_v2). Structure determines "whether this thing is knowable."

What result would refute Prediction 2: if on the structure-broken experimental group, enlarging parameters 5–10-fold stably restores performance to the same level as the unbroken group (the error gap disappears to insignificance), then "structural information cannot be compensated by parameters" is falsified—meaning large models can hard-learn the broken structure through capacity, and structure's independent causal standing does not hold.

Prediction 3 (structure correct, substrate replaceable)

Statement: as long as structure S remains isomorphic, switching the implementation substrate from one to another (carbon-based molecules $\rightarrow$ silicon circuits, biological neurons $\rightarrow$ artificial neurons, simulated agents $\rightarrow$ real robots) can preserve the system's core function.

UDOS measured prototype (reports7/splat_transfer_demo.json, cpu-proto): when the geometric structures of Real-to-Sim and Sim-to-Real are consistent, the robust-transfer strategy has success rate 1.0 and collisions 0.0, with significant margin (margin = 0.22) over naive transfer's 0.5/0.5. This is a small but real piece of evidence that "spatial geometric structure is preserved across real/simulated substrates."

What result would refute Prediction 3: if there exists a class of repeatedly verified tasks whose core function strongly depends on the physical properties of a particular substrate (for example a cognitive function that only emerges under the specific diffusion time constant of wet chemical synapses and cannot be reproduced in silicon electronic circuits no matter how structure is adjusted), then "cross-substrate isomorphism" is falsified, and this paper must retreat to the narrower claim that "structure is transferable only within a certain substrate family."

Prediction 4 (the threshold of structure prior to learning: before structure reaches the threshold, gains from adding data/parameters tend to zero)

Statement: there exists a topology/structure threshold. Before structure reaches that threshold, the marginal gains from increasing data volume and parameter count rapidly tend to zero; only once structure crosses the threshold is the parameter–data scaling activated. Structure is the "fourth axis" of scaling.

UDOS measured prototype (reports7/agent_scaling.json, $k \leq 2$ verified, $k > 2$ cpu-proto): ensemble quality saturates with agent count k as $\text{mse}(k) = a + b/k$, with fit $a = 0.009252$, $b = -4.23 \times 10^{-5}$; when newly added members are only near-correlated copies of the strongest expert (structurally introducing no new connections/information sources), $b \approx 0$ and gains go to zero. This echoes the "fourth structure axis" beyond Chinchilla's "parameters and data grow in proportion."

What result would refute Prediction 4: if on a topology clearly "wrong" structure (for example randomly connecting all experts into a fully connected noise graph, or changing a layered structure into a flat layer with no interfaces), simply enlarging parameters and data in Chinchilla proportions still lets performance keep falling along a power law with no saturation inflection, then the "structure threshold" is falsified—meaning structure is only a proxy for parameters, not an independent axis.

The common refutation logic of the four predictions: this paper's central claim C is killed by the stable reproduction of any one of "topology does not affect capability," "structure can be compensated by parameters," "substrate is not replaceable," or "structure has no threshold." These four are not rhetoric; they are this paper's epitaph conditions.

3 Natural-history evidence (I): "structure determines function" at four scales

This chapter traverses four natural domains by a unified four-step template: **what structure is → how structure determines function → counterexamples of breaking/changing structure → isomorphic mapping to AI**. These four steps are the methodological master template of the whole paper, and Chapter 4 reuses the same template for UDOS evidence.

Figure 2 Isomorphic comparison table of the four natural-history domains

Figure 2 Comparison of the "structure → function → counterexample" chains across the four natural domains. Each row's structure differs, but the abstracted causal shape is consistent: components unchanged, only structure changes, and function changes. Schematic.

3.1 Chemical bonds and materials: isomerism is the cleanest natural experiment

What structure is. Starting from the quantum-mechanical explanation of chemical bonds, Pauling, in the first 1939 edition of *The Nature of the Chemical Bond*, systematically established a causal chain: chemical bonds determine molecular structure, and molecular structure determines material behavior (Linus Pauling research archive, Lindau Science Library; verified 2026-09-19). Bond types (ionic, covalent, metallic), the spatial arrangement of the crystal lattice, and atomic positions within the unit cell together determine a material's macroscopic properties: graphite and diamond, both pure carbon, one soft and conductive, the other hard and insulating, differing only in whether the carbon atoms are arranged in layers or tetrahedrally bonded. This is the most direct example of "connectivity topology + geometric organization" determining physical properties.

How structure determines function. The basic belief of materials science is: macroscopic properties (hardness, conductivity, melting point, reactivity) are not a direct function of "what elements a material is" but of "how those elements are arranged." The same iron, arranged body-centered cubic versus face-centered cubic, has very different toughness and strength.

Counterexamples of breaking/changing structure—*isomerism*. The key evidence of this section is isomerism. Structural isomers have exactly the same molecular formula (the same atom types and counts, i.e. the same "components/parameters"), and merely because the connection order of atoms differs, they become different substances with very different physical and chemical properties. Taking butane $\mathrm{C_4H_{10}}$: n-butane is a straight chain and isobutane (2-methylpropane) is branched, with different boiling points; positional isomerism moves a functional group one position along the carbon chain and reactivity changes; functional-group isomerism turns an alcohol (e.g. ethanol $\mathrm{C_2H_5OH}$) and an ether (e.g. dimethyl ether $\mathrm{CH_3OCH_3}$), same formula $\mathrm{C_2H_6O}$ into two completely different classes of compounds. This is the cleanest controlled experiment in nature: **components identical, parameters identical, the only variable is connection structure, and the result is vastly different function**. It directly supports the "under the same components structure determines function" side of central claim C.

Isomorphic mapping to AI. This is fully isomorphic to "isomerism" in model design: two neural networks with the same parameter count and training data, merely because of different connection topologies (dense connections vs sparse routing, residual vs direct, layered vs flat), can differ order-of-magnitude in capability and efficiency. The fan-in difference of UDOS star vs layered in Section 4.2 below is the multi-agent version of "n-butane vs isobutane."

The diamond-and-graphite example must also be added, because it exposes especially clearly that "geometry/spatial organization $\$G\$$ also independently determines function." Diamond and graphite are both pure carbon, and their atomic connection topology (carbon atoms bonding to each other) is almost identical stoichiometrically; the difference is only in how the carbon atoms are arranged in 3D space: in diamond each carbon bonds tetrahedrally to four neighbors, forming a rigid 3D network, so it is hardest and non-conductive; in graphite carbon is arranged in in-layer hexagonal grids, with layers stacked only by weak van der Waals forces, so it is soft, serves as a lubricant, and conducts along the layer direction. **The same material, almost the same bonding, merely because the 3D arrangement (geometric organization $\$G\$$) differs, the physical properties are as different as cloud and mud.** This example is especially important to this paper: it shows that even fixing $\$T\$$ (connections) and changing only $\$G\$$ (spatial arrangement), function changes drastically— $\$G\$$ is not a vassal of $\$T\$$; it is an independent dimension in the triple. Section 4.3 below, "multi-view pose structure determines novel-view prediction," is precisely the engineering counterpart of $\$G\$$ acting independently.

3.2 Carbon chains and life: structural organization precedes special substances

What structure is. Carbon's tetravalence lets it form chains and rings and build skeletons; the spatial arrangement of functional groups on the skeleton determines a molecule's chemical behavior. Life is possible not because the element carbon has anything mysterious—silicon, nitrogen, and phosphorus can also bond—but because the structural organization of "carbon tetravalence forming chains + functional groups being replicable and mutable" happens to support an information-storage and replication mechanism that is "heritable, mutable, evolvable."

How structure determines function. Watson and Crick's 1953 Nature paper A Structure for Deoxyribose Nucleic Acid (Nature 171(4356):737–738, 1953-04-25; awarded the 1962 Nobel Prize in Physiology or Medicine) revealed: DNA's replication ability comes directly from its double-helix structure—two complementary strands wound by A-T, C-G base pairing, with one strand's sequence uniquely determining the other. The paper's famous line "it has not escaped our notice" says precisely: once the structure is seen, the replication mechanism is foreseen. **It is structure itself, not what rare material DNA uses, that makes heredity possible.**

Counterexamples of breaking/changing structure. Changing carbon-chain structure without changing components makes life collapse or mutate: reversing the spatial orientation of a chiral center (enantiomers), mirror molecules often have very different pharmacological activity in the body—one enantiomer is a medicine and the other may be toxic (the thalidomide event is the classic lesson of chiral-drug tragedy; its precise pharmacological details belong to accepted medical history and this paper does not expand exact doses). This shows: even mirror-flipping one sign in 3D geometry changes function drastically—geometric organization $\$G\$$ is no optional decoration.

Isomorphic mapping to AI. This chapter's implication for AI is profound: **life originates from structural organization, not some special substance.** Extrapolating to intelligence: if intelligence is also a "structural organization," then in principle it can be realized on any substrate that can support such relations—this is precisely the natural-history basis of Prediction 3 (substrate replaceable). Carbon-based is no more a privilege of intelligence than it is a privilege of life.

3.3 Proteins: sequences fold into shapes, and shape is function

What structure is. Proteins have four levels of structure: the primary is the linear amino-acid sequence, the secondary is local folds such as the α -helix/ β -sheet (Pauling, Corey, and Branson correctly proposed the α -helix in 1951, the spring-like structure of about 3.6 residues per turn maintained by intra-chain

hydrogen bonds), the tertiary is the 3D folded shape of the whole peptide chain, and the quaternary is how multiple peptide chains assemble. Pauling, for his contributions to chemical bonds and molecular structure, was called by Crick one of the "fathers of molecular biology."

How structure determines function. Anfinsen's denaturation–renaturation experiments on ribonuclease proved: all the information needed for a protein's 3D active conformation is already contained in its primary amino-acid sequence—placing the denatured protein back under appropriate conditions makes it spontaneously fold back into the active native conformation. This is called the Anfinsen principle (thermodynamic hypothesis), and Anfinsen was awarded the 1972 Nobel Prize in Chemistry, with his summary lecture *Principles that Govern the Folding of Protein Chains* published in *Science* 181:223–230 (1973, PubMed 4124164). Function is determined by "shape": an enzyme's catalysis relies on the geometry of its active site (the lock-and-key model and the later induced-fit model), and allostery relies on a structural change at one site transmitted through the folded shape to a distant site. **The protein does not "know" what it is catalyzing; what its binding site looks like determines what it grabs.**

Counterexamples of breaking/changing structure—denaturation. Protein denaturation is the cleanest counterexample of "structure broken, components unchanged": heating or adding urea breaks hydrogen bonds, and not one of the peptide chain's primary sequences (amino-acid connection order and composition) is lost, but the 3D fold comes apart and activity is lost. The elegance of Anfinsen's experiment lies precisely here—he reversibly broke apart and restored this structure, proving information is in structure and function relies on structure. More recent counter-evidence comes from AlphaFold: Jumper et al. 2021 in *Nature* published Highly accurate protein structure prediction with AlphaFold (*Nature* 596(7873):583–589, DOI:10.1038/s41586-021-03819-2), reaching median backbone accuracy 0.96 Å r.m.s.d.95 at CASP14. **Predicting structure is predicting function**—because as long as the shape is right, binding and catalytic functions are almost locked in. This is the strongest engineering confirmation in the AI era of the proposition "shape = function."

Isomorphic mapping to AI. The protein's "sequence → fold → function" chain is highly isomorphic to the neural network's "weights → connection graph → behavior" chain: the former has information in the sequence and function in the folded shape; the latter has parameters in weights and function in connection geometry. AlphaFold uses pure computation to predict folding from sequence, just as we hope pure computation can predict an agent system's coordination capability from connection topology. Prediction 2's "breaking structure means losing function" is denaturation for proteins; "adding parameters cannot compensate" is for proteins "lengthening the peptide chain by ten thousand amino acids cannot restore the already-scattered active-site shape."

The two mechanisms of "shape determines function" also need to be stated more precisely, otherwise they are easily read as a slogan. First is **lock-and-key and induced fit**: an enzyme's active site is not a passive pocket but undergoes local conformational change upon substrate binding (induced fit)—i.e. function comes from the geometric matching of site shape and substrate shape, with binding specificity almost entirely determined by "whether the geometry fits." Second is **allostery**: when a distant site binds a small molecule, it transmits the signal to the active site through geometric distortion of the whole folded chain, changing its activity. Allostery is the ultimate "structure transmits function"—information crosses segments not through new chemical bonds but only through the chained deformation of 3D shape. Together these two show: a protein's "intelligence" (selective recognition, remote regulation) is written entirely in its geometric structure, not on any "computing unit." The reason AlphaFold's revolution shocked all of biology is that it turned the "sequence → structure" step computable—and once structure is predictable, the whole functional chain of binding, catalysis, and

allostery becomes inferable. This paper carries this logic to AI: if "structure $\rightarrow$ function" holds for proteins, then for agent systems, "connection topology $\rightarrow$ coordination capability" should likewise be computable and predictable.

3.4 Neurons and the brain: intelligence lies not in the unit but in the connectome

What structure is. Cajal, through silver staining around 1888, proved: the nervous system is made of discrete neurons (the "neuron doctrine," the term named by Waldeyer in 1891, the word synapse named by Sherrington), and neurons transmit signals through contact regions (synapses) rather than forming a continuous net; Cajal shared the 1906 Nobel Prize in Physiology or Medicine with Golgi, who held the "reticular theory." The revolutionary point of this discovery is: **it rewrote "what the nervous system is" from a continuous mass of jelly into a graph of discrete nodes connected by specific connections.** The location of intelligence was, for the first time, precisely pointed to "how the nodes connect."

How structure determines function. Hebb, in *The Organization of Behavior* (New York: Wiley, 1949), proposed: when neuron A's axon repeatedly and persistently participates in firing neuron B, A's transmission efficiency to B strengthens—later generations condensed this idea into "neurons that fire together wire together," forming a "cell assembly": a group of neurons repeatedly co-activated forms a structure as the physical carrier of an idea or memory. Note that "fire together, wire together" is a later mnemonic condensation, and Hebb's own original statement is a more cautious statement about synaptic-efficiency change. This mechanism shows: **on the biological side, learning is essentially shaping connection structure**—dynamics (firing rules) in turn shape structure (connection weights and patterns).

Counterexamples of breaking/changing structure—connectome lesions and spatial encoding. Breaking connections while preserving cells makes function lost: in brain injury, even if surviving neurons themselves are intact, as long as key connection bundles are cut, the corresponding cognitive function is lost—direct counter-evidence that "intelligence lies in connections, not units." The spatial-cognition example is especially precise. Tolman in 1948 in *Psychological Review* 55(4):189–208 (DOI: 10.1037/h0061626) proposed the "cognitive map": a rat running a maze does not memorize a string of stimulus–response actions but internally builds a spatial-structure representation of the environment, so it can take shortcuts never traveled. In 1971 O'Keefe discovered "place cells" in the hippocampus—specific neurons fire when the rat reaches a specific location; in 2005 May-Britt Moser and Edvard Moser discovered "grid cells" in the entorhinal cortex—firing positions tile a periodic hexagonal grid in the environment, forming an internal coordinate system. O'Keefe and the Mosers were awarded the 2014 Nobel Prize in Physiology or Medicine. **The brain encoded "spatial structure" itself into neuronal connections and firing patterns**—this is the hardest landing in neuroscience for this paper's title "spatial structure is itself intelligence."

Isomorphic mapping to AI. This chapter's mapping to AI is all-around: the Transformer's self-attention is a "learnable connection topology" (Section 4.1); the hippocampus's place/grid cells prove that for a system to navigate in space it must explicitly encode "positional relations" into structure—mutually confirming the decisive role of UDOS Chapter 4's multi-view pose structure on novel-view prediction (Section 4.3). Hebb also reminds us of an important bidirectionality: learning shapes structure, and structure also constrains learning—this must be honestly handled in the "structure is both cause and effect" boundary of Chapter 6.

Two more neuron-side pieces of evidence are worth adding, because they directly support this paper's "structure" stance. First is **connectomics**: a century after Cajal proposed the neuron doctrine, modern

connectomics systematically maps complete neural connection diagrams, with the basic belief—to understand what a nervous system is doing, knowing what neurons exist and how each fires is insufficient; one must know the complete connection diagram among them. This is fully consistent with this paper's claim that "intelligence lies in connections, not units." Second is **criticality**: many experimental and theoretical works point out that living neural networks seem to spontaneously self-organize into a "critical state"—neither quiet nor bursting, where signals can just propagate through the system without losing control. If this picture holds, then "whether structure organizes the system to criticality" is an independent structural-design problem, not a mere parameter-adjustment problem. This paper does not write these two as settled conclusions (connectomics' functional mapping remains early, criticality's experimental evidence remains disputed), only using them as evidence that "structure is seriously treated as an independent research object in neuroscience." When precise original literature is needed, please supplement dedicated reviews of connectomics and neural criticality [CITATION NEEDED].

Natural-history summary: from chemical bonds to neurons, the same causal shape repeatedly appears at four scales—**components unchanged, only structure (topology/geometry/modules) changes, and function changes; breaking structure, even with parameters and energy present, function is lost.** Table 1 converges this shape into a table.

Table 1 The four natural-history chapters' "structure → function → counterexample → AI mapping"

Scale	Protagonist of the structure triple	Function determined by structure	Counterexample of "components unchanged, structure changed"	Isomorphic mapping to AI
Chemical bonds/ materials	Bond type + lattice geometry	Macroscopic physical properties	Isomerism (same formula, different connections, very different properties)	Capability difference of same-parameter different-topology networks
Carbon chains/life	Carbon tetravalence chains + functional-group positions	Replicable and mutable	Enantiomers (chirality flipped, pharmacology very different)	Structure can cross substrates (Prediction 3)
Proteins	Primary sequence → 3D fold	Binding/catalytic function	Denaturation (chain unbroken, fold scattered, activity lost)	Predicting structure is predicting function (AlphaFold)
Neurons/ brain	Connectome + hierarchical modules	Memory/spatial navigation	Connection-bundle lesions (cells alive, function lost)	Attention = learnable connection topology

4 AI and UDOS evidence (II): making structure a measurable engineering quantity

This chapter is the original-evidence core of this paper. Unlike the natural-history evidence of Chapter 3, the figures here come from the CPU deterministic prototype of UDOS Reasoning Engine v7.5.0 (commit 1a71270 series), all tracing back to the raw JSON in reports7/, marked by evidence level. The meaning of

evidence levels must first be stated, lest figures be misread as "already verified on real large models/real machines":

- **verified**: results reproducible on fixed-seed deterministic CPU and with key assertions passing contract tests;
- **cpu-proto**: point estimates on a CPU prototype, with the mechanism shape credible but constants not calibrated across multiple seeds and real machines;
- **cpu-proxy**: replacing real processes with synthetic proxies (e.g. `asyncio.sleep` simulating remote I/O, fixed-seed perturbation simulating real noise);
- **unverified**: external industry figures or media retellings, used only as background, not mutually verified.

General rules for experiments and ablations (AlexNet-style "cost of removing X"). The four core measurements of this chapter follow one discipline: **whenever claiming "some structure matters," one must give the "cost of removing/breaking that structure," not only report "adding it is better."** The way AlexNet proved depth cannot be deleted is "removing any convolutional layer $\approx -2\%$ top-1"; the counterparts in this chapter are: (a) P1 star vs layered—at the same $\$N\$$ the star center's fan-in explodes (infeasible) while layered presses to constant 9, this being "the cost of removing layered structure (retreating to star)"; (b) P8 single-view vs multi-view—removing pose geometry $\$G\$$, IoU falls from 0.8897 back to 0.7147; (c) P7 pure-semantic/pure-motion vs layered—removing modular boundaries $\$M\$$ (no handoff), direct contact tasks 0%, motion ordered-sorting 0%; (d) P6 short window—removing observation structure, acceleration skill= -0.508 (worse than the mean). Each group is read by **"experimental setup $\rightarrow$ comparison $\rightarrow$ ablation/sensitivity $\rightarrow$ validity threat"**: base/seed/data volume are given below, the comparison is the same-budget baseline, the ablation removes some structural variable, and validity threats are concentrated in Chapter 9.

4.1 Transformer / Attention: the content–position–relation sequence of structural priors

What structure is. The Transformer established by Vaswani et al. 2017 in Attention Is All You Need (NeurIPS 2017, arXiv:1706.03762) essentially transforms "connection topology" from the RNN's temporal hard connections and the CNN's local hard connections into a **learnable connection topology**: self-attention lets any two positions in a sequence interact directly, with connection weights computed dynamically from content. There is an often-overlooked structural hierarchy here—content (token representation), position (positional encoding), and relation (attention matrix) are three different things, and the attention matrix is a learnable graph of "who relates to whom and how strongly."

How structure determines function. Self-attention's success is not because it has "more parameters" but because it changes long-range-dependency connections from "must pass through intermediate layers one by one" to "any two points directly connected"—precisely a change in connectivity topology $\$T\$$. MoE (Mixture-of-Experts) further pushes "trading structure for capacity" to the extreme: Fedus, Zoph, and Shazeer's Switch Transformer (JMLR 2021, arXiv:2101.03961) uses top-1 sparse routing, making total parameters about 1.6T and training about $7\times$ faster, but each forward activates only a small fraction of experts—**total parameters did not increase; the routing structure of "which parameters are connected into this computation" changed.** This is the textbook case of "trading structure for capacity": parameter count is appearance, routing structure is what is designed.

Position relative to this paper's claim. Transformer and MoE together show: a significant part of modern large-model progress occurs on the "structure axis," not the pure parameter axis. This provides the first engineering-side support for the "fourth structure axis" of Chapter 8.

It is worth unpacking the three layers of "content–position–relation" clearly, because they correspond exactly to this paper's structure triple. In self-attention, for a token to be understood, three types of information are needed: content (who this token itself is, provided by embedding), position (where it is in the sequence, provided by positional encoding), and relation (what connections and how strong between it and every other token in the sequence, provided by the attention matrix). These three are not the same: you can fix content and position (the same sentence, same order) and only recompute the attention relation matrix, and the model's behavior changes; you can also fix attention relations (which two tokens attend to each other) and only change content representation, and behavior changes again.

The relation layer (attention matrix) is the connectivity topology \$T\$ this paper speaks of, except it is learnable and dynamic. This is extremely important to this paper: it shows modern large models did not discard "structure" but turned "connectivity topology" from hand-designed hard connections into soft connections dynamically determined by content—structure did not disappear, it was learned parameterically. This is in principle the same as Hebb's "co-firing shapes connections": connection strength is determined by usage history. Section 6.4 below, when discussing "structure is both cause and effect," takes attention as the most modern example. MoE's sparse routing designs structure in another direction: it does not let connections be dynamically learned but pre-designs on a static architecture the topology of "which tokens should be routed to which experts"—the convergence of hand-designed and learnable structural design.

4.2 P1: changing only topology, coordination capability differs by six orders of magnitude (direct evidence for Prediction 1)

This is the core measurement of Prediction 1. Data are taken from reports7/matrix_scale_demo.json (evidence level cpu-*proto*; the first two levels 10, 100 are explicit simulations and consistent with closed-form formula values *explicit_matches_formula*=true, the latter four levels 1,000 to 1,000,000 are analytical closed-form extrapolations).

Under exactly the same task (the layered coordination matrix of 24 work orders), changing only the connection topology from "star" to "b-ary layered tree," two structural quantities are measured: center fan-in (the number of messages the center node must receive instantaneously) and critical-path rounds (the message round-trip steps needed to complete coordination). Results are in Table 2.

Table 2 UDOS topology six levels: star vs layered (only \$T\$ changed, parameters and tasks unchanged)

Target agent count	Actual unit count	Depth	Star center fan-in	Layered maximum fan-in	Star rounds	Layered rounds	Evidence/ level
10	73	2	640	9	256	4	explicit / cpu-proto
100	585	3	5,120	9	2,048	6	explicit / cpu-proto
1,000	4,681	4	40,960	9	16,384	8	analytical / cpu-proto
10,000	37,449	5	327,680	9	131,072	10	analytical / cpu-proto
100,000	299,593	6	2,621,440	9	1,048,576	12	analytical / cpu-proto
1,000,000	2,396,745	7	20,971,520	9	8,388,608	14	analytical / cpu-proto

Figure 3 UDOS topology fan-in/rounds double-logarithmic chart

Figure 3 Horizontal axis = target agent count N (log scale, $10 \rightarrow 10^6$), left vertical axis = center/maximum fan-in, right vertical axis = critical-path rounds. Red dots: star explodes linearly ($\beta=1$) with N (at $N=10^6$ fan-in 20,971,520, rounds 8,388,608); cyan/purple lines: layered presses fan-in to constant 9 and rounds to log level (at $N=10^6$ only 14 rounds). Baseline = same N and task volume, only topology T differs. Data trace to matrix_scale_demo.json; evidence level: cpu-proto (first two levels explicit, latter four analytical extrapolations).

Reading the figure: at million-level scale, the star center node must simultaneously receive about 20.97 million messages (center fan-in), and completing coordination needs about 8.39 million critical-path rounds; the layered tree makes each node's maximum fan-in constantly 9 (except the root, with 8 subordinate units), and the critical path needs only 14 rounds. **Parameter count, task volume, and energy budget are exactly identical; the only variable is connectivity topology T , and coordination overhead differs by about six orders of magnitude.** This is the direct prototype of Prediction 1 in this repository.

It must be honestly stated: the P1 paper itself also notes this repository did not measure "the collapse comparison of a pure star under center failure," so the fault-robustness side of "layered outperforms star" remains a planned experiment; Table 2 measures the structural observables of **coordination complexity (fan-in, rounds)**, not end-to-end task success rate. Translating fan-in/rounds into real latency and success rate needs measurement in a real distributed environment—this is the GPU/real-machine gate of Chapter 7.

4.3 P8: multi-view pose structure determines novel-view prediction (evidence for geometry G)

What structure is. Novel-view synthesis solves: from images of known views, predict what an object looks like from a never-seen view. The structural variable here is **camera geometry and multi-view pose constraints**—i.e. the relative pose relations of observation views in 3D space (geometry/spatial organization G).

Measurement. Data are taken from reports7/spatial_novelview_demo.json (v7.3.8, cpu-proto, TSDF explicit voxel representation, 8 training views, 6 held-out views, voxel 0.1m). Comparing "single view" with "multi view with pose structure":

Table 3 P8: the effect of multi-view pose structure on novel-view prediction (cpu-proto)

Metric	Single view	Multi view (with pose structure)	Change
Contour IoU	0.7147	0.8897	+0.1750
Coverage	0.7456	0.9957	+0.2501
Precision	0.9472	0.8931	-0.0541
Depth MAE (m)	0.3186	0.0924	-0.2262

Reading the figure: merely adding the "relative pose structure of multiple views" (parameters and training data essentially unchanged), held-out views' contour IoU rises from 0.7147 to 0.8897, coverage from 0.7456 to 0.9957, and depth error falls to about one-third. The report's own conclusion is: "multi-view spatial context's contour IoU/coverage on held-out views is significantly higher than single view, proving novel-view prediction relies on camera geometry and 3D consistency."—precisely direct evidence that geometry/spatial organization $\$G\$$ determines prediction capability, isomorphic to the hippocampus's place/grid cells encoding spatial relations into structure (Section 3.4). splat_transfer_demo.json (v7.3.9, cpu-proto) supplements: the CPU prototype of 150 3D Gaussians reaches novel-view IoU 0.7285 and coverage 1.0, with Sim-to-Real robust-transfer success 1.0/collisions 0.0 (vs naive transfer 0.5/0.5, margin 0.22)—spatial geometric structure preserved across simulation/reality, a small engineering piece of evidence for Prediction 3.

4.4 P7: the layered structure of cognitive layer $\times$ motion priors, 4.7% takeover rate for 100% success (evidence for modular boundaries $\$M\$$)

What structure is. In embodied control there have long been two schools: general large models are strong at semantic planning but weak at dense contact control; motion priors (diffusion policies etc.) are strong at high-frequency contact but weak at semantic order. P7's structural scheme is **layering**: the high-level semantic layer only scores the low-level motion-prior candidates when necessary, choosing between "adopt the prior" and "issue a short corrective takeover." This is a design of modular boundaries $\$M\$$ and levels—encapsulating "cognition" and "motion" into two independently replaceable modules, handing control by confidence.

Measurement. Data are taken from the P7 paper abstract and its source reports7/embodied_hybrid_demo.json (v7.3.7, cpu-proto, paired experiments of 4 tasks $\times$ 10 rounds, 40 rounds per mode):

Table 4 P7: control mode $\times$ success rate/takeover rate/tokens (cpu-proto, pilot point estimate)

Mode	Aggregate success rate	Mean score	Collisions	LLM calls	Takeover rate	Token proxy (in/out)
Pure semantic direct	75%	76.97	0.25	Every decision segment	100%	616k / 182k
Pure motion prior motion	52.5%	69.57	0	Zero	0%	0 / 0
Layered hybrid	100%	100	0	Only when necessary	4.7%	331k / 40k

Reading the figure: any single module alone is incomplete—direct is 0% on contact-door tasks, motion is 0% on ordered-sorting tasks; but once the two are assembled by modular boundaries through a **layered structure**, hybrid simultaneously gains both sides' capabilities, reaching 100% success and zero collisions with only 4.7% takeover (i.e. 95.3% of decision segments resolved by motion priors themselves), with token proxy volume also significantly falling. Task decomposition shows takeovers highly concentrated on tasks where "the semantic layer is truly irreplaceable." This is direct evidence that modular boundaries $\$M\$$ determine system capability: **not which module is stronger, but how you layer and hand off between them determines whether the whole can succeed**. It shares origin with SayCan (Ahn et al., 2022, arXiv:2204.01691)'s two-layer structure of "LLM gives step candidates $\times$ skill value function gives feasibility."

4.5 P6 counterexample: when structural information is insufficient, even adding parameters cannot identify (hard counterexample for Prediction 2)

This section is the most important reverse-direction evidence of the whole paper and the most direct support for Prediction 2: **structure determines "whether this thing is knowable."**

Data are taken from the estimator_param_error of reports7/v7_verification.json (v7.0.3, main gate verified, estimator part cpu-proto). The system-identification task must reverse-infer hidden dynamics parameters from observation data. Results are as follows:

Table 5 P6: structure (observation window/observability) determines parameter identifiability (verified/cpu-proto)

Parameter to identify	MAE	Mean-prediction MAE	Identifiability skill	Mean observability	Evidence
Initial velocity v_0	0.6606	0.9683	0.318	0.771	cpu-proto (n=480)
Acceleration a (structure insufficient)	0.9256	0.6138	-0.508	0.798	cpu-proto (n=160)
Spring frequency ω	0.1471	0.2563	0.426	0.914	cpu-proto (n=160)
Other v_2	0.0432	0.2396	0.820	0.908	cpu-proto (n=160)

Reading the figure: a negative identifiability skill means "our estimate is worse than directly predicting with the global mean." The acceleration parameter a 's skill is **-0.508**—because under the observation structure of a short window (window=6) it is **fundamentally unidentifiable**: although mean observability is not low (0.798), structurally acceleration cannot be observed from the given window. **At**

this point no matter how many parameters or model capacity you add, you cannot conjure information fundamentally not contained in the observation structure. This is precisely the reverse confirmation of Prediction 2: when structure (here observation-window/observability structure) is missing, parameter compensation fails. The P5 world model under oracle structure blind test overall MSE falls from v701's 0.0459 to 0.01037, with all conformal coverage passing (the marginal coverage interval of the six-combination merge 0.7758–0.9577, nominal 0.8/0.9/0.95), showing once structure and the data flywheel are in place, capability is truly released.

The theoretical implication of this result is worth expanding, because it touches the more general proposition "structure determines knowability." System identification has a classic and cold conclusion: **whether a parameter can be estimated from data depends not on how strong a model you use to fit, but on whether your observations structurally "excite" that parameter's influence**—the core of observability and identifiability theory. If the observation structure (window length, sensor configuration, kinematic constraints) itself does not let acceleration leave distinguishable traces, then no matter how wide or deep you make the neural network, it can only fit the mean, and skill is naturally negative. UDOS's P6 measurement (skill=−0.508) is precisely a small engineering verification of this theory. Translating into this paper's language: **information availability is a structural property, not a parameter property.** This is fully consistent with "breaking structure cannot be compensated by adding parameters" (Prediction 2)—you cannot use a larger model to recover a degree of freedom that never appears in the observation structure. This is also why this paper repeatedly stresses "design structure right first": on a wrong observation/connection structure, more parameters only fit noise and the mean.

4.6 Other structural modules: handoff bundle, quorum, market conservation, state coverage, Stigmergy

The UDOS matrix lands structural principles on multiple mechanism modules; here each gets one set of measured figures as material for the master table.

- **P10 handoff bundle structure** (udos7/topology/transfer.py, agent_tool.py): handoff is not forwarding but responsibility transfer, with a fixed envelope + six-field validation; if validation fails the Owner does not change and responsibility does not slip away with the message. This is "the interface contract of modular boundaries $\$M\$$ "—making cross-module handoff a structured, verifiable bundle rather than unstructured text forwarding.
- **P2 BFT-lite quorum structure** (udos7/topology/consensus.py): committee size $\$n \geq 3f+1\$$, quorum $\$2f+1\$$, same-round equivocation invalidates the whole round. At $\$n=7, f=2\$$ (the minimum legal committee with $\$3f+1=7\$$), the 7-person QA committee under 2 Byzantine counter-voting members correctly accepts all 24 work orders (24/24), with the faulty fleet triggering 5 reworks and isolating 3 faulty agents. The "structure" here is voting topology and majority threshold—it directly reuses the topology conclusion of classic BFT (Castro & Liskov, OSDI 1999, $\$f < n/3\$$).
- **P11 market conservation structure** (udos7/topology/market.py): the internal task market's budget is a conserved quantity. In matrix integration budget = paid = balance_sum = 240 (24 work orders $\times$ 10/order), conserved=true. Conservation is a structural invariant—it does not depend on which specific agent did which thing, only on the accounting structure of "where money comes from and goes."
- **P4 state-coverage structure** (reports7/egodata_flywheel_demo.json, v7.4.0, cpu-proto): the quality of active data collection depends on the coverage structure of the "state space." Oracle state units 226; passive collection covers only 157 (oracle ratio 0.695, yield 0.6667, redundancy 0.599); coverage-aware active collection covers 215 (oracle ratio 0.951, yield 0.8542). **Replacing passive**

data-piling with structured planning of "which state units to cover" raises oracle coverage from 0.695 to 0.951—this is the structure of the data flywheel, not data volume.

- **Stigmergy environmental-medium structure** (udos7/topology/stigmergy.py, v7.4.10, cpu-proto): agents do not directly negotiate but atomically claim tasks and leave completion marks on a shared blackboard, letting the environment bear memory. On the 30-task/6-agent benchmark, stigmergy uses 2 traces per task for **60 messages total**; contract-net (broadcast–bid–award) uses about 8 negotiation messages per task for **240 total**. **Changing the "communication structure" from point-to-point negotiation to "environment bearing memory" reduces messages to about one quarter**—the result of jointly designing connectivity topology \$T\$ and modular boundaries (the environment as a shared module), isomorphic to Dorigo's ant pheromones (1996).

4.7 Master table: natural structure ↔ UDOS modules ↔ measured figures ↔ evidence levels ↔ common abstraction

Table 6 Cross-chapter master table (this paper's evidence convergence)

Natural-structure case	UDOS corresponding module	Measured figures (trace reports7)	Evidence level	Common abstraction
Isomerism: components identical, structure differs	P1 star vs layered topology	Fan-in 20,971,520→9; rounds 8,388,608→14	cpu-proto	Same parameters, only \$T\$ changes, capability order-of-magnitude difference
Protein geometry (shape = binding)	P8 multi-view pose structure	IoU 0.7147→0.8897; coverage 0.7456→0.9957	cpu-proto	\$G\$ (pose geometry) determines prediction
Brain layering/modularization	P7 cognition × motion layering	100% success; takeover 4.7%	cpu-proto pilot point estimate	\$M\$ (layered handoff) determines overall performance
Denaturation: structure lost, function lost	P6 short-window unidentifiability	Acceleration skill = −0.508	cpu-proto (main gate verified)	Structure missing, parameters cannot compensate (Prediction 2)
Cell assembly/Hebb co-connection	P9 ensemble saturation $mse=a+b/k$	$a=0.009252$, $b=-4.23e-5$; equal-weight bad expert 0.2131	$k \leq 2$ verified / $k > 2$ cpu-proto	Homogenization (no new structure) expansion saturates
Connectome/synapse topology	P2 BFT-lite quorum	$n \geq 3f+1$, 24/24 accepted, 3 faulty isolated	cpu-proto	Voting topology determines safety/liveness
Conservation (metabolic structure)	P11 market settlement	budget=paid=240 conserved	cpu-proto	Structural invariant does not depend on individuals
Spatial cognitive map	P4 state-coverage flywheel	Oracle coverage 0.695→0.951	cpu-proto	Coverage structure determines data value
Ant stigmergy	P1 blackboard-medium collaboration	60 vs contract-net 240 messages	cpu-proto	Environment bears memory, reducing point-to-point negotiation
Cross-substrate (carbon/silicon)	P8/P8-splat Sim2Real	Robust transfer 1.0/0 collisions, margin 0.22	cpu-proto	Geometric structure preserved across simulation/reality (Prediction 3)

Reading Table 6: every row says the same thing—measuring "structure" independently from parameters reveals its decisive contribution to system capability. The natural-history side (Chapter 3) and the engineering side (this chapter) meet end-to-end in this table.

5 Unified framework: the structure–dynamics–energy ternary model and cross-substrate isomorphism

5.1 The ternary model

Tightening the distinctions of Chapter 2 into a working model: any intelligent system is a triple

$$\text{System capability} = f(S, D, E) \tag{1}$$

where $S=(T,G,M)$ is structure (this paper's protagonist), D is dynamics (learning rules, signal propagation, energy conversion), and E is the energy budget. The three **multiply rather than add**—if any one approaches zero, system capability approaches zero:

- Only S without D : a circuit diagram drawn on paper, a dead brain, a denatured peptide chain—structure present but not "alive," function zero;
- Only D without S : a randomly stirred chemical soup, an unstructured fully connected noise network—dynamics present but relations not organized, unable to produce predictable coordination;
- E insufficient: structure and dynamics both right, but energy cannot be supplied, the brain falls into coma, the model underfits.

The whole claim of this paper, while acknowledging this multiplicative relation, stresses that **when D and E are controlled (fixed), the marginal contribution of S has been systematically underestimated**. Chapter 6 repeatedly returns to this model to handle its counterexamples.

5.2 The theorem-like statement of cross-substrate isomorphism

Compressing the evidence of Chapters 3 and 4 into a testable "theorem-like statement" (note: this is not a mathematical theorem but a research-program statement awaiting rigorous proof):

Substrate-Isomorphism Conjecture: suppose task τ has a "structure-sufficient solution" S^* . If two substrates $\mathcal{M}_1$ (e.g. carbon-based biology) and $\mathcal{M}_2$ (e.g. silicon machines) can both realize the triple $S^*=(T^*,G^*,M^*)$ at sufficient resolution, and dynamics D and energy E fall within the feasible interval of task τ , then the two substrates' core functional performance on τ can be isomorphic. Substrate differences manifest in constants (speed, energy, precision), not in the presence or absence of function.

The value of this statement lies in its **clearly drawn falsifiable boundary**: it does not claim any substrate can run any structure (that is strong functionalism, which this paper does not endorse), only that "when structure is sufficiently realized and dynamics/energy are in the feasible interval," function can transfer across substrates. Prediction 3 is one testable version of this proposition. Section 6.3 discusses why "random structure also works" does not necessarily refute this proposition—because the proposition requires a "structure-sufficient solution," not "any structure."

5.3 The structure axis: the fourth axis omitted by scaling law

Existing scaling literature writes performance as a function of parameters N , data D , and compute C (Kaplan 2020 power law; Hoffmann/Chinchilla 2022 corrected to N and D in proportion). This paper does not deny this three-axis curve but argues it implicitly freezes structure. UDOS's P9 measurement (agent_scaling.json) shows: when newly added agents are only near-correlated copies of the strongest expert—i.e. structurally introducing no new connections/information sources—ensemble quality saturates as $mse(k)=a+b/k$, with the saturation floor a determined by the best single expert's quality ($a=0.009252$, close to $best_single=0.009196$). This shows: **when structure (independence among information sources/connection diversity) does not change, simply increasing "head count" (parameters/members) gains tend to zero**. This does not contradict Chinchilla's "parameters and data grow in proportion" but fills the omitted dimension: structure diversity. On the structure axis, "expansion" is not the more the better, but depends on whether you

really introduce new connection structure—homogeneous expansion (structure unchanged) saturates, heterogeneous expansion (structure changed) keeps improving.

Figure 4 Conceptual structure–parameter–performance surface

Figure 4 Conceptual surface schematic. Horizontal axis parameters/compute, vertical axis the structure axis (topology maturity), upright axis intelligent performance. Before structure crosses the topology threshold (red threshold line), marginal gains from adding parameters tend to zero; after crossing, parameter scaling is activated. Schematic, not measured data.

6 Counter-arguments and boundaries: what this paper does not claim

The honesty of a flagship paper is shown in whether it fully writes the counter-arguments. This chapter lists serious objections to central claim C one by one, along with our responses and concessions.

6.1 Structure is not a sufficient condition: dead brain, denatured protein, empty topology

The most direct objection is: **structure is only necessary, not sufficient**. An anatomically complete dead brain, connectome intact and neurons all present, but without metabolism and electrical activity (\$D\$ and \$E\$ zero), has no intelligence. A denatured protein, primary sequence (connection order) unchanged, only the 3D fold scattered—this instead shows "complete structure" requires geometry \$G\$ to also be present. A neural network with only a connection graph but no signals or learning rules, no matter how large its parameters, does nothing. **This paper fully accepts these counterexamples:** they precisely confirm the ternary multiplicative model of Section 5.1—structure is one multiplier, not the whole. This paper never claims "with structure there is intelligence," only that "with dynamics and energy fixed, structure is the dominant variable."

6.2 The functionalism vs substrate-neutrality debate

"Substrate replaceable" (Prediction 3) directly steps on the old debate in philosophy of mind between functionalism and biological naturalism. One side (strong functionalism) claims intelligence is entirely independent of substrate; the other (biological naturalism/embodied school) claims intelligence essentially depends on specific physical properties of biological wetware. This paper's stance is a moderate, testable middle ground: we do not take a metaphysical stance, only giving the falsifiable version of Prediction 3—"in task families where structure is sufficiently realized and dynamics/energy feasible, substrate differences manifest in constants." Section 2.4 already states what would refute it: if some cognitive function strongly depends on the specific time constant of wet chemical synapses and silicon cannot reproduce it, this paper retreats to "structure is transferable only within a substrate family." **We converted a metaphysical stance into an experimental question.**

6.3 Random structure also works: reservoir computing and the lottery hypothesis

A serious technical objection: randomly connected networks can also work. Reservoir computing / echo-state networks use random fixed recurrent connections and only train the output layer to complete time-series prediction; the lottery ticket hypothesis claims dense randomly initialized networks contain sparse "winning" substructures. If random structure works, how can one still say "structure is dominant"?

This paper's response is two-layer. First, the reservoir's "random" is random within a carefully designed structural family—the spectral radius of recurrent connections is tuned to criticality (guaranteeing the echo-state property), itself a deep structural constraint; it is not arbitrary randomness but "randomness designed to be critical." Second, the lottery hypothesis precisely supports this paper: it says "training filters a randomly initialized network into a winning substructure"—i.e. **the learning process keeps useful connection structures and prunes useless ones**, precisely evidence of structure being shaped. That random structure works does not mean structure is unimportant—it only shows good structures can be searched out. This instead leads to the next boundary.

6.4 Learning itself shapes structure: structure is both cause and effect

Hebb's "co-firing strengthens connections" reminds us: in biological neural networks, structure (connection weights and patterns) is itself a product of learning. This brings a causal-direction trouble—**structure is both the cause of intelligence and the result of learning**. How does this paper handle this loop? Our stance is: this does not refute the central claim, only refines it. Within a single system's lifecycle, learning shapes structure; but in cross-system comparison (Prediction 1: star vs layered) and cross-task design (Prediction 4: threshold), structure is an independently designable, independently measurable independent variable. UDOS's P1 experiment measures structural effects precisely under the controlled condition of "learning rules identical, only connection topology changed"—this controlled design temporarily separates "structure shaping" from "structure effects." Honestly, **the bidirectional coupling of structure and learning is a simplification of this paper's framework**, and complete unification requires bringing learning dynamics $\$D\$$ into the closed loop of structure evolution, an open problem of Chapter 7.

6.5 Description granularity: when "structure" degenerates into an unfalsifiable all-purpose explanation

This paper must self-guard against an epistemological trap: **the word "structure" is too useful, so useful it may become an all-purpose basket that explains everything and therefore explains nothing**. If any performance difference can be post-hoc attributed to "different structure," central claim C becomes unfalsifiable. This paper's defense has three parts: first, Section 2.1 pins structure to the triple $\$S=(T,G,M)\$,$ each item with operationalizable measurements (fan-in, pose, module interfaces), not a vague "way of organizing"; second, all four predictions state explicit refutation conditions (Section 2.4), and this paper will be killed by specific experimental results; third, every row in the UDOS master table (Table 6) gives concrete figures and evidence levels, not allowing the word "structure" to cover unmeasured black boxes. **When "structure" cannot be reduced to any item of T/G/M nor give refutation conditions, it is an explanation that should be rejected**. This self-constraint, more than any positive argument, protects this paper from degenerating into metaphysics.

6.6 External validity of this repository: CPU prototypes are not LLMs, not real machines

Evidence levels must be stated thoroughly; this is the landing point of this paper's academic honesty. All UDOS measurements are **Linux CPU deterministic prototypes**: the semantic judge is deterministic rules rather than a real vision-language model; the environment is a point-mass double-integrator agent rather than MuJoCo-MJX contact simulation or a real machine; tokens are proxy accounting rather than real API billing; remote agents' I/O curves are simulated by `asyncio.sleep` rather than real LLM calls. The P9 paper itself also states: "all figures are CPU fixed-seed pilot point estimates, without multi-seed

confidence intervals; their value lies in mechanism shape and failure boundaries, not precise constants." Therefore:

- The six-order-of-magnitude fan-in difference of Table 2 is **deterministic mathematics** (closed form consistent with explicit simulation), its "shape" credible; but translating it into real distributed systems' latency and success rate is the GPU/real-machine gate of Chapter 7;
- P7's 100% success is a pilot point estimate under a **synthetic point-mass environment**, not constituting a SOTA claim about real VLA models;
- P6's skill=-0.508 is measured on a **synthetically injected defect structure**, and real-world "structure missing" is more complex.

The claim strength of all engineering evidence in this paper does not exceed its respective evidence level. This is not modesty but the requirement of falsifiability—a proposition that dares not mark its evidence boundaries does not deserve to be called science.

6.7 Relation to scaling law: not denying, only adding an axis

Finally clarifying the relation to existing scaling literature to avoid being misread as "anti-scaling." This paper **does not deny** the Kaplan/Chinchilla parameter–data–compute three-axis curve; on the contrary, this paper treats it as a premise. This paper claims: those three axes were measured under "structure fixed" conditions—all use the Transformer, all stack parameters by established topology. Once structure is also made a design variable, a fourth axis appears (Figure 4). The test method this paper gives is the simple **same-parameter same-compute comparison**: fixing parameter count, compute, and data, changing only connection topology/module boundaries, to see whether performance jumps. If it does, structure is an independent axis; if not, this paper's Prediction 1 is falsified. This is not overthrowing scaling law but extending it from three dimensions to four.

7 Testable research program: the gate roadmap from CPU prototypes to GPU/real machines

This chapter turns the four predictions into executable experimental plans, each marked with a gate and current blocker.

GPU gate for Prediction 1: in a real multi-GPU distributed environment, using the same batch of real LLM agents (not CPU synthetic agents), run end-to-end task success rates and wall-clock latency for star vs layered topology, confirming whether the "six-order-of-magnitude fan-in difference" translates into "order-of-magnitude differences in success rate/latency." Current blocker: needs LLM keys and cloud budget (AL4 gate), which CPU prototypes cannot bear.

Parameter-compensation gate for Prediction 2: on the structure-broken experimental group (random rewiring, scrambled poses), enlarge parameters 5–10-fold to see whether performance recovers. Current blocker: real large-model parameter-expansion training needs GPU and compute budget; UDOS currently has only the 967,796-parameter world model (v7.0.3, verified), insufficient for large-scale parameter-compensation experiments.

Real-machine gate for Prediction 3: migrate the spatial geometric structure of P8/P8-splat from the CPU Gaussian-splat prototype to real depth-camera calibration, multi-view capture, and MuJoCo-MJX contact simulation, verifying "geometric structure preserved across simulation/reality." Current blocker: full 3DGS training and real-time rendering need GPU, real camera calibration and multi-view capture need hardware, real-machine Sim-to-Real needs GPU/HPC.

Structure-threshold gate for Prediction 4: under Chinchilla-style compute budgets, systematically scan the fourth dimension of "topology maturity," verifying whether a structure threshold exists—below it gains from parameters/data tend to zero, above it activated. Current blocker: needs large-scale training compute and multiple topology comparisons, far beyond CPU-prototype capability.

Table 7 The gate roadmap of the four predictions

Prediction	Current evidence (CPU prototype)	Gate needed to unlock	Falsification observation
P1 topology trades for capability	Fan-in/rounds six levels (cpu-proto)	Real LLM agents + cloud GPU	Star/layered success same order of magnitude
P2 broken structure cannot compensate	Acceleration skill= -0.508	Large-scale parameter-expansion training	Adding $5\text{--}10\times$ parameters restores performance
P3 substrate replaceable	Sim2Real 1.0/0 (cpu-proto)	Real machine + MuJoCo-MJX	Some function strongly depends on wetware, non-transferable
P4 structure threshold	$\text{mse}=\text{a}+\text{b}/\text{k}$ saturation (verified/cpu-proto)	Large-scale topology $\times$ compute scan	On wrong structure still falls along power law

8 Discussion: architecture design, embodiment, multi-agent, and AGI

8.1 Implications for architecture design

If this paper's framework holds, then the architect's first-principles action should not be "choosing how large a model" but "designing the correct structure": how to topologize connections (avoiding center fan-in explosion), how to organize space (explicitly encoding pose and geometry), how to layer modules (cognition and motion, planning and execution handing off by confidence). UDOS's P1 (layered decoupling of fan-in), P7 (cognition $\times$ motion layering, 4.7% takeover), and P10 (handoff bundle as responsibility interface) jointly point to one engineering principle: **design structure right first, then discuss parameter scale**. This is in line with the post-Chinchilla industry awareness that "parameters are not the bigger the better," only pushing that awareness from "parameter-data ratio" to "structure itself."

More concretely, this principle has three operational implications for daily engineering decisions. First, **the fan-in budget should be written into design documents before the parameter budget**: if a center node must instantaneously receive tens of thousands of messages, no matter how large a model hangs behind it, it becomes a system bottleneck—P1 turned this intuition from "seems like a traffic jam" into computable fan-in/rounds upper bounds. Second, **geometry and pose should be first-class citizens, not attached byproducts of images**: P8 proved explicitly encoding multi-view relative pose can raise novel-view coverage to 0.9957; this means in embodied and 3D-system design, "how cameras are placed, how coordinates are aligned" is no preprocessing triviality but a structural variable determining the ceiling. Third, **module interfaces should have contracts like handoff bundles**: P10 and P7 both show cross-module handoff, once made into unstructured text forwarding, dilutes responsibility in forwarding; making handoff a verifiable, responsibility-bound bundle lets the system safely layer and scale. The common foundation of these three implications is the same belief—structure is an engineering object that can be designed, budgeted, and verified, not a byproduct emerging only after training.

8.2 Implications for embodied intelligence

Embodied intelligence's deepest lesson comes from neuroscience: the brain encoded spatial structure (place cells, grid cells) into connection patterns (Section 3.4). UDOS's P8 multi-view pose structure (Table 3) and P4 state-coverage flywheel (oracle coverage 0.695→0.951) reproduce the same principle on the engineering side—**embodied systems must explicitly encode "spatial relations and state coverage" into structure, not expect large models to figure it out themselves from raw pixels**. P7's layered hybrid control (Table 4) then shows: the embodied three-layer structure of "perception–planning–control" is no arbitrary engineering layering but a mirror of the biological cognitive architecture (hippocampus–cortex–cerebellum).

8.3 Implications for multi-agent systems

Multi-agent legions' most common narrative is "the more heads the stronger," but UDOS's P9 measurement (agent_scaling.json) gives a hard boundary: quality saturates as $\$mse(k)=a+b/k\$$, equal-weight averaging with one bad expert drags MSE to 0.2131 (far worse than calibrated weighting's 0.0092); throughput peaks at 2 concurrency 244.172 tasks/s then falls (4/8 concurrency down to 158.4/106.5). **The "expansion" dividend of multi-agent depends on whether you introduce new connection structure (information independence), not head count**. Combined with P1's layered topology, P2's BFT quorum, P11's market conservation, and Stigmergy's blackboard medium, UDOS gives a construction diagram of "how multi-agent should design structure": layering reduces fan-in, quorum ensures safety, market conservation manages incentives, blackboard reduces negotiation traffic.

8.4 Restrained implications for AGI

This paper's implications for AGI must be extremely restrained, avoiding sliding into big words. If "intelligence is a structural organization transferable across substrates" (Prediction 3), then a question worth studying is: **is there a structure isomorphic to carbon-based neurons but efficiently implemented on silicon?** This paper does not predict when AGI arrives nor claims structure alone leads to AGI. It only provides a research stance: rather than infinitely stacking parameters on a fixed structure (that axis's marginal gains already show saturation signs in P9), treat "finding the correct structure" itself as a searchable, measurable engineering problem. If AGI arrives, it will not be "a larger same network" but more likely "a new structure not yet designed"—this is the only, and restrained, hint this paper's framework gives to AGI research.

9 Threats to validity and limitations

This section is this paper's independent validity-threat chapter. Chapter 6 handles "counter-argument philosophy" (whether structure is sufficient, the substrate-neutrality debate); this section, standing in the reviewer's position, lists the engineering-evidence threats by the five categories of internal/external/construct/statistical/causal—more honest than NSA (with no independent Limitations section in the whole text), and more reviewable than only mentioning "cpu-proto" in scattered paragraphs.

9.1 Internal validity: ablation purity and whether the "removal cost" is really caused by structure

- **Ablations are structural observables, not end-to-end success rates:** Table 2 measures fan-in/rounds (deterministic mathematics), not real task success rates. The P1 paper itself already stated "did not measure the collapse comparison under pure-star center failure," so the "layered is more robust" side remains a planned experiment.
- **Single-seed pilots:** P7's 100% success, P8's IoU improvement, and P6's skill=-0.508 are all single-seed, without 95% confidence intervals. **Needs ≥ 30 seeds before submission.**
- **Actively disclose biases adverse to oneself** (AlexNet footnote-2 honesty): the latter four levels of Table 2 (1,000→1,000,000) are analytical closed-form extrapolations, not explicit simulations; only the first two levels (10, 100) have explicit_matches_formula=true. Writing extrapolation levels as "measured" is exaggeration, and this paper strictly marks them.

9.2 External validity: CPU prototypes are not LLMs, not real machines

UDOS is all Linux CPU deterministic prototypes: the semantic judge is rules not a VLM, the environment is a point-mass double-integrator, tokens are proxy accounting, I/O is asyncio.sleep. Table 2's six-order-of-magnitude difference is deterministic mathematics (shape credible), but translating into real distributed systems' latency and success rate is Chapter 7's GPU/real-machine gate; P7's 100% success is a pilot point estimate under a synthetic environment, not constituting a SOTA claim about real VLA.

9.3 Construct validity: "structure" must be reducible to T/G/M, otherwise it degenerates into an all-purpose explanation

Section 6.5 already self-guarded: if any performance difference is post-hoc attributed to "structure," the central claim becomes unfalsifiable. This section reiterates the defense: each item of the triple $S=(T,G,M)$ has operationalizable measurements (fan-in/pose/module interfaces), all four predictions have refutation conditions, and each row of the master table has figures and evidence levels. **When a "structure difference" cannot be reduced to any item of T/G/M nor give refutation conditions, it is an explanation that should be rejected.**

9.4 Statistical validity: single-point small samples must not be written as settled conclusions

None of the four core measurements has error bars/confidence intervals; P6's skill=-0.508 is measured on a synthetically injected defect structure (n=160), and real-world "structure missing" is more complex. This paper does not write any difference as a statistically significant conclusion; all differences are marked "pilot point estimates."

9.5 Causal validity: structure is both cause and effect

Hebb and the lottery hypothesis point out structure can be shaped by learning; this paper's controlled comparisons measure "initial structure" effects, not "structure evolving with scale during training." The complete closed loop requires bringing learning dynamics $\$D\$$ into structure-evolution tracking—already listed as a Chapter 7 open problem.

9.6 Validity-threat master table

Threat type	Most likely attack	How this paper defends	Unresolved ledger
Internal	Ablations measure fan-in not success rate	Deterministic-mathematics marking, no latency claim	GPU end-to-end comparison
External	cpu- <i>proto</i> $\neq$ real machine	Four-level evidence, no SOTA claim	Real-machine gate
Construct	"Structure" becomes all-purpose explanation	Triple operationalization + refutation conditions	—
Statistical	Single seed no error bars	Pilot-point-estimate marking	≥ 30 seeds/95% CI
Causal	Structure shaped by learning	Controlled-comparison separation	In-training structure-evolution tracking

10 Conclusion

This paper argued a restrained, falsifiable central proposition: given fixed components and energy budget, a system's intelligent behavior is primarily determined by its spatial relational structure $S=(T,G,M)$, and structure is the invariant across substrates.

The natural-history road (Chapter 3) shows: from chemical-bond isomerism and carbon-chain chirality flips, to protein folding and denaturation, to neuronal connectomes and grid cells, four scales repeatedly stage the same causal chain—**components unchanged, only structure changes, and function changes**. The engineering road (Chapter 4), through UDOS measurements, made this causal chain a measurable quantity: star vs layered topology pressed coordination overhead by six orders of magnitude, multi-view pose raised novel-view coverage to 0.9957, layered hybrid control with 4.7% takeover reached 100% success, and when structure was missing adding parameters was to no avail (skill=-0.508).

This paper did not overstate. Chapter 6 fully acknowledges: structure is not a sufficient condition, dead brains and denatured proteins remind us structure $\times$ dynamics $\times$ energy are all indispensable; random structure also works; learning itself shapes structure; this repository's CPU prototypes have limited external validity. This paper's boundaries are strictly constrained by the four predictions' refutation conditions—they are this paper's epitaph.

Closing to the title: **spatial structure is itself intelligence**. This sentence is no flattering rhetoric but an engineering instruction: measure structure, design structure, treat structure as the fourth axis alongside parameters, data, and compute. When a system shows intelligence, do not rush to praise its material or its training volume—first ask what its spatial relations look like.

References

Note: the following AI/systems literature is reused from this volume's Master Reference Library (verification date 2026-09-19, status , and this paper does not rewrite the library itself; newly added natural-science classics were verified one by one on 2026-09-19 using `general_search/web`, and those that could not be verified are marked [CITATION NEEDED], strictly forbidding fabricated DOI/years.

A. Natural-science classics (newly added in this paper, verified 2026-09-19)

1. **Watson, J. D. & Crick, F. H. C.** (1953). Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid. **Nature**, 171(4356), 737–738. (Double helix and complementary base pairing; 1962 Nobel Prize in Physiology or Medicine, shared with Wilkins; Rosalind Franklin and Gosling's X-ray diffraction was key evidence.) Verification sources: NIH Profiles in Science / NHGRI / Nature 171:737–738 bibliographic record. ✓
2. **Anfinsen, C. B.** (1973). Principles that Govern the Folding of Protein Chains. **Science**, 181(4096), 223–230. (Anfinsen principle/thermodynamic hypothesis: amino-acid sequence determines tertiary structure; ribonuclease denaturation/renaturation; 1972 Nobel Prize in Chemistry.) PubMed: 4124164. Verification sources: official Nobel Prize page / Science 181:223–230 bibliography. ✓
3. **Pauling, L.** (1939). The Nature of the Chemical Bond and the Structure of Molecules and Crystals. Cornell University Press. (Chemical bonds determine molecular structure, molecular structure determines behavior; proposed the α -helix with Corey and Branson in 1951.) Verification sources: Lindau Nobel Mediatheque / Berkeley Graduate Lectures. ✓
4. **Ramón y Cajal, S.** (silver-stain observations from about 1888; 1906 Nobel Prize in Physiology or Medicine, shared with Golgi). Neuron doctrine: the nervous system is made of discrete neurons transmitting signals through synaptic contact regions ("neuron" named by Waldeyer 1891, "synapse" by Sherrington). Verification sources: official Nobel Prize 1906 page / Scholarpedia Cajal entry. ✓
5. **Hebb, D. O.** (1949). The Organization of Behavior: A Neuropsychological Theory. New York: Wiley. (Co-firing strengthens synaptic efficiency, cell assemblies; "fire together, wire together" is a later mnemonic condensation, not the author's original words.) Verification sources: Frontiers history of synaptic plasticity / multiple neuroscience textbooks retelling the original sentence. ✓
6. **Jumper, J., Evans, R., Pritzel, A. et al.** (2021). Highly accurate protein structure prediction with AlphaFold. **Nature**, 596(7873), 583–589. DOI:10.1038/s41586-021-03819-2. (CASP14 median backbone accuracy 0.96 Å r.m.s.d.-95.) Verification sources: EMBL-EBI citation page / Nature 596:583–589. ✓
7. **Tolman, E. C.** (1948). Cognitive maps in rats and men. **Psychological Review**, 55(4), 189–208. DOI: 10.1037/h0061626. (Cognitive map: the rat's internal representation of the environment's spatial structure, not a stimulus-response chain.) Verification sources: Psychological Review 55(4):189–208 bibliography. ✓
8. **O'Keefe, J.** (1971). Discovery of hippocampal place cells; **Moser, M.-B. & Moser, E. I.** (2005). Discovery of entorhinal grid cells. They shared the **2014 Nobel Prize in Physiology or Medicine** with O'Keefe. Verification source: official Nobel Prize 2014 press release. ✓ (Note: this paper did not open the precise journal volume/pages of O'Keefe's 1971 original paper and the Mosers' 2005 Nature paper one by one; when precise citation is needed in writing, use the official Nobel page.)
9. Isomerism/structural isomerism: same molecular formula, different atomic connection order, different physical/chemical properties (e.g. n-butane vs isobutane; ethanol vs dimethyl ether $\text{C}_2\text{H}_6\text{O}$). Verification sources: organic-chemistry textbooks/encyclopedias general statements. ✓ (Textbook-level common knowledge, no single original source needed.)

B. AI and systems literature (reused from Master Reference Library, 2026-09-19 ✓)

1. **Vaswani, A. et al.** (2017). Attention Is All You Need. NeurIPS 2017. arXiv:1706.03762.
2. **Hoffmann, J. et al.** (2022). Training Compute-Optimal Large Language Models (Chinchilla). NeurIPS 2022. arXiv:2203.15556.

3. **Kaplan, J. et al.** (2020). Scaling Laws for Neural Language Models. arXiv:2001.08361.
4. **Fedus, E., Zoph, B. & Shazeer, N.** (2021). Switch Transformer. JMLR 2021. arXiv:2101.03961.
5. **Castro, M. & Liskov, B.** (1999). Practical Byzantine Fault Tolerance (PBFT). OSDI 1999.
6. **Smith, R. G.** (1980). The Contract Net Protocol. IEEE Trans. Computers, C-29(12). DOI:10.1109/TC.1980.1675516.
7. **Dorigo, M., Maniezzo, V. & Colorni, A.** (1996). The Ant System. IEEE Trans. SMC-B, 26(1).
8. **Amdahl, G. M.** (1967). Validity of the Single Processor Approach to Achieving Large Scale Computing Capabilities. AFIPS SJCC 1967.
9. **Ahn, M. et al.** (2022). Do As I Can, Not As I Say (SayCan). arXiv:2204.01691.
10. **Mildenhall, B. et al.** (2020). NeRF: Neural Radiance Fields. ECCV 2020. arXiv:2003.08934.
11. **Kerbl, B. et al.** (2023). 3D Gaussian Splatting. SIGGRAPH 2023. arXiv:2308.04079.
12. **Curless, B. & Levoy, M.** (1996). A Volumetric Method for Building Complex Models from Range Images (TSDF). SIGGRAPH 1996. DOI:10.1145/237170.237269.
13. **Krogh, A. & Vedelsby, J.** (1995). Neural Network Ensembles, Cross Validation, and Active Learning. NeurIPS 1995.
14. **Yin, M. et al.** (2019). HotStuff. PODC 2019. arXiv:1803.05069.
15. **Wu, Q. et al.** (2023). AutoGen: Multi-Agent Conversation Framework. arXiv:2308.08155.

C. Unverified placeholders (forbidden to use as real references)

- [CITATION NEEDED] : precise authors/years of the original reservoir-computing (echo-state network) literature and the original lottery-ticket-hypothesis literature; precise journal volume/pages of O'Keefe 1971 and the Mosers 2005 original papers—this paper only cites their Nobel conclusions, with precise sources pending retrieval verification.

Appendix A: UDOS measurement reproduction index

All figures trace to release-v5.4.4/udos-engine/reports7/ and udos7/topology/. Reproduction commands, seeds, and evidence levels follow the appendices of the topical papers (P1–P12).

Figure cited in this paper	Source file	Field	Evidence level
Star fan-in 20,971,520 / rounds 8,388,608; layered fan-in 9 / rounds 14	matrix_scale_demo.json	clean.scale[*]	cpu-proto
Multi-view IoU 0.7147→0.8897, coverage 0.7456→0.9957	spatial_novelview_demo.json	results.single_view/multi_view	cpu-proto
3DGS prototype IoU 0.7285; Sim2Real 1.0/0.5	splat_transfer_demo.json	gaussian_splat/sim_to_real	cpu-proto
Hybrid 100%, takeover 4.7%	P7 abstract + embodied_hybrid_demo.json	aggregate	cpu-proto pilot point estimate
Acceleration skill=-0.508	v7_verification.json	estimator_param_error.accel_a	cpu-proto (main gate verified)
mse=a+b/k, a=0.009252, b=-4.23e-5	agent_scaling.json	quality_saturation_fit	k≤2 verified / k>2 cpu-proto
Equal-weight bad-expert MSE 0.2131	agent_scaling.json	quality_curve[*].naive_equal_2_mse	verified
State-coverage oracle 0.695→0.951	egodata_flywheel_demo.json	passive/coverage_aware_collection	cpu-proto
Stigmergy 60 vs contract-net 240 messages	P1 paper + udos7/topology/stigmergy.py	contract_net_message_count	cpu-proto
Market budget conservation 240	matrix_scale_demo.json	clean.market.conservated	cpu-proto
World model 967,796 parameters, blind overall 0.01037	v7_verification.json	params/test_metrics.blind	verified

(End of full text. This paper is the UDOS Reasoning Engine v7.5.0 whole-volume flagship synthesis paper P0, mutually verified with the P1–P12 topical papers; detailed reproduction steps for all UDOS figures are in the topical papers' appendices.)

Trilogy interlinks: this paper is the "structure axis." The prologue is P0c Metacognition: To Know What One Knows and What One Does Not Know — Engineering AI Self-Cognition and Boundary Calibration (self-boundary axis); the scale axis is P0b Structure + Scale, Everything Follows Scaling Laws.